

DWDM① Data Mining

a) Data mining is the process of discovering useful patterns, relationship and knowledge from large data sets using statistical.

Two Data Mining Tasks

\* classification    \* clustering.

Steps in Data KD

\* Data cleaning → Remove noise and missing values.

\* Data integration → combine data from multiple.

Role of Data preprocessing

\* improves data quality.

\* Remove inconsistency and redundancy.

\* increase mining accuracy.

b) Data cleaning:- Data cleaning is the process of detecting and correcting errors, missing values, duplicate.

Two Data quality issues:-

\* Missing values    \* Noisy data.

Handling missing values:-

\* Ignore the record

Example:-

Age values 20, 21, Null, 23 → Replace Null with Mean  $(21.3 \approx 21)$

② a) Data Integration:- It is the process of combining data from multiple source into a single, consistent dataset.

Challenges:-

\* Data redundancy \* Data inconsistency.

Handling redundancy.

\* Remove duplicate records \* correlation analysis

Handling inconsistency:-

\* standardize formats \* Resolve conflicting values.

b) Covariance analysis

covariance measures how two variables change together.

Population covariance

$$\text{cov}(x, y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

sample covariance:

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

③ Covariance interpretation

a) Given

$$A = (2, 3, 4, 5, 6)$$

$$B = (5, 8, 10, 11, 14)$$

$$\text{Mean}(A) = 4 \quad \text{Mean}(B) = 9.6$$

Products:-

$$(-2)(-4.6) = 9.2$$

$$(-1)(-1.6) = 1.6$$

$$(1) (0.4) = 0.4$$

$$(0) (1.4) = 0$$

$$2(4.4) = 8.8$$

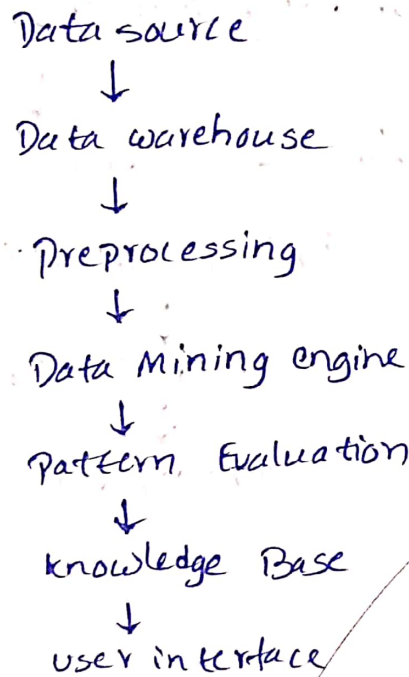
Total = 20

$$\text{Sample Covariance} = \frac{20}{4} = 5$$

$$\text{Covariance} = 5 \text{ (positive)}$$

Covariance is positive, the stocks, not independent.

## b) Architecture of Data Mining system



## 4) Find Covariance

a) Given  $x = (4, 5, 6, 7, 9)$   $y = (8, 3, 7, 5, 6)$

$$\text{Mean}(x) = 6.2$$

$$\text{Mean}(y) = 5.8$$

Products:-

$$(-2.2)(2.2) = -4.84$$

$$(0.8)(-0.8) = -0.64$$

$$(-1.2)(-2.8) = 3.36$$

$$(2.8)(0.2) = 0.56$$

$$(-0.2)(1.2) = -0.24$$

$$\boxed{\text{Sum} = -1.8}$$

$$\text{Sample Covariance} = -\frac{1.8}{4}$$

$$= -0.45$$

b) Major steps in Data preprocessing

\* Data cleaning    \* Data integration

\* Data transformation    \* Data Reduction

\* Data Discretization.

Pipeline

collect data  $\rightarrow$  Handling missing values



standard format  $\leftarrow$  Remove outliers

Normalize data  $\rightarrow$  Reduce Dimensions



Discretize Data



Ready for Data Mining.

⑤  
a)

Given data (30 values)

$$\text{Bin size} = \frac{30}{3}$$

$\therefore$  10 values per bin

Bin 1:-

13, 15, 16, 16, 19, 20, 20, 21, 22, 22. Mean = 18.4

smoothed Bin:-

18.4, 18.4, 18.4, 18.4, 18.4, 18.4, 18.4, 18.4, 18.4, 18.4. Mean = 18.4

Bin 2:-

25, 25, 25, 25, 30, 33, 33, 35, 35, 35. Mean =  $\frac{30+1}{2} = 15.5$

smoothed:- 15.5 repeat 10 times.



Bin 3:- 35, 36, 40, 45, 46, 52, 70.

Mean = 46.3

Replace each by 46.3

b) Data Discretization:-

Data Discretion converts continuous numerical values into intervals.

importance:-

- \* Reduce data complexity.

- \* Remove noise.

equal width Binning

- \* All bins have equal range.

- \* simple and fast.

Equal frequency

- \* Handles skewed data better.

- \* Bin widths may vary.

